

The Copyright protection about digital images based on web

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Abstract—The emphasis and difficulty of developing a photography website contains not only a whole design of the site and digitization conversion and repair of papery photos but also copyright protection of the image that is the site builder must focus on. In order to avoid malicious tort and leak, we classified images and controlled browsing and downloading permissions of the classified pictures by using different techniques before web design, and we have already achieved good effect in the image copyright protection.

Keywords—copyright protection; digital image; photography website construction; watermark

I. INTRODUCTION

Through the construction of photography pictures website based on the network environment, picture file resources are able to be saved effectively, and the cost of storage picture files can be reduced greatly. At the same time, if only the network platform is properly designed, the convenience of using pictures resources can be improved greatly by using appropriate web design technology and adding pictures searching function for web.

The emphasis and difficulty of developing a photography website contains not only a whole design of the site and digitization conversion and repair of papery photos but also copyright protection of the image that is the site builder must focus on. In order to avoid a malicious tort and leak, we classified images and controlled browsing and downloading permissions of the classified pictures by using different techniques before web design (In Table 1), and we have already achieved a good effect in the image copyright protection.

TABLE1. PICTURE TYPE, AUTHORITY AND TECHNICAL PLAN

Picture Type	Authority	Technical Plan
Picture Type	can be consulted and downloaded by ordinary visitors	Pictures add watermark
Retained pictures	can be consulted by ordinary visitors, but can't be downloaded	FLASH technology
Confidential pictures	can be showed and downloaded by a particular user	Membership system

Now the research result is to be introduced.

II. ADDING DIGITAL WATERMARK ON GENERAL PICTURE AND DECLARING PICTURES COPYRIGHT

Adding digital watermark is the effective method to protect copyright. It embed in multimedia data copyright information such as numbers, serial number, text, images signs, in order to do the copyright protecting and communication security, identification of data file and product label, etc [1]. There is visible watermark and invisible watermark according to whether the watermark signal is visible and not visible [2]. The main function of visible watermark is clearly declaring copyright. Previewed pictures on most webs are embedded by the visible watermark. The most common electronic watermark technology to protect network pictures is to add a copyright tag by the software on a specific area of the picture that is to be uploaded [3]. The tag will become the inseparable part of the original data.

Adding watermark on pictures to declare copyright is the first work before the pictures are uploaded. Because the quantity of images needed to upload is enormous, we experimented by different software to test the process efficiency. We added watermark on pictures and uploaded them to the server by different software that includes Photoshop CS in Plan 1, Snag It 9 in Plan 2 and AspJpeg plug-in in Plan 3. The test procedure is in Figure 1.

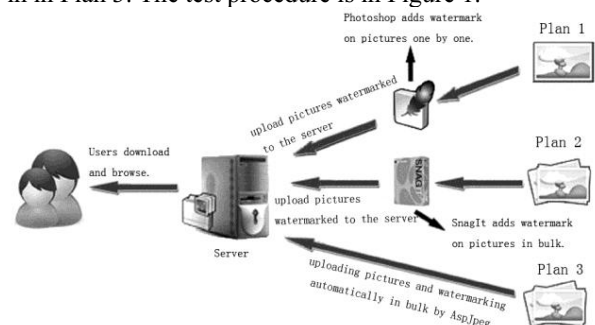


Figure 1. The flow chart of using different software to add digital watermark

In Plan 3, AspJpeg is a powerful image processing components based on Microsoft IIS. It can dynamically create images of high quality watermark and miniatures in the ASP/ASP.Net by using few codes. It placed the finished

watermark on the server, then it installed AspJpeg plug-in in the server and it write corresponding codes into the uploaded components in the website. Then in the process of uploading the pictures the server automatically executed to the commands of watermark for the image. It finished adding watermark and uploading at the same time so that the uploading efficiency can improve greatly. It can also generate miniatures so that amount of work decreased greatly. We list the results to compare in Table 2.

TABLE 2. THE PROCESS, EFFICIENCY AND EFFECTIVENESS OF DIFFERENT SOFTWARE ADDING DIGITAL WATERMARKING

Serial number	Software	Implementation process	Efficiency	Effect
1	Photoshop CS	Photoshop added watermark on pictures one by one, then uploads finished pictures in bulk.	2pic/ min	Efficiency is low, workload is big. But the watermark position can be controled freely to avoid affecting image ensemble.
2	SnagIt 9, Batch Picture Protector etc.	This type of software could add watermark on pictures in bulk, then uploaded finished pictures.	5-10pic/ min	Efficiency is higher, workload is smaller. But the watermark position can't be controled freely so that effect of a part of picture is affected.
3	AspJpeg plugin	This plugin was installed in the server. It could add watermark on pictures being uploaded. The efficiency improved greatly.	20-30pic/ min	Efficiency is high, workload is small. But the watermark position can't be controled freely so that effect of a part of picture is affected. At the same time the web designer is requested higher.

According to the test results, we choose Plan 1 and Plan 3. Most of pictures were added watermark by AspJpeg plug-in and the minority that was easily affected by watermark was processed by Photoshop. Good results have been achieved. However, adding digital watermark is generally used to only confirm copyright when the copyright disputes happen. This is only an afterwards protective measure. And the thief often can remove image watermark by technology, so the network image copyright infringement problems will not be solved only by adding watermarks in pictures.

III. PREVENT THE RESERVE IMAGES BE DOWNLOADED BY USING FLASH TECHNOLOGY

Many websites offer the users some forums to publish photography, and some users hope their images can't be downloaded but displayed.

According to this request, we made several trials. First, we used JavaScript language in web page to forbid the users right click and copy the pictures. For example, we write these codes between <head> and </ head>:

```
< SCRIPT LANGUAGE = JavaScript id= client Event
Handlers JS>
<!-- Function document _ on mouse down ( ) {
If (event button 2)
{Alert ("All rights reserved, forbidden copying"); // this
dialog appear when the user use the right button of mouse
return false ;}
}
```

```
Function document _ on select start ( ) {
Alert ("Shenyang normal university is the copyright
owner!"); // this dialog appear when the user select all
return false ;}
} /-->
</ SCRIPT>
< SCRIPT language e= JavaScript event = on mouse down
for= document>
<!-- Document _ on mouse down ( ) /-->
<!--copy forbid function, end-->
</ SCRIPT>
< SCRIPT language e= JavaScript event= on select start
for= document>
<!-- document _ on select start ( ) /-->
</ SCRIPT>
```

These codes mean: when the users use the right button of mouse, the message box "All rights reserved, forbidden copying" will appear; when the users select all, the message box "Shenyang normal university is the copyright owner!" will appear.

Thus, we can effectively prevent the copyright infringement. At the same time we have also expressed the statement of copyright reserved.

In tests we found that although we have forbidden the right button of mouse, many web pages appeared this button of "save the pictures in this image" because the users have installed some special software. So, this method to protect the copyright is easily cracked.

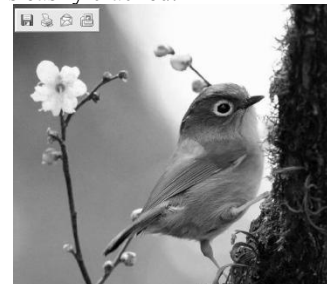


Figure 2. For bidden the copy function, but the picture will automatically appear "Save this image" button

The tests proved that the pictures inevitably were saved by the users only if they are showed in web page. So we made bold hypothesis: Can we avoid this problem by converting images into other formats? For Flash is very popular in web now, we first consider converting images into SWF, and it can avoid naked images be displayed in web pages.

But, it is unrealistic to convert thousands of pictures into SWF in Flash. In practice, we designed a player which contains buttons of zooming, forward and backward, etc by using the function of 'loadmovie'. Then the following steps were processed:(1)We divided the pictures into different folders according to the different author, and the naming way of each picture is same. For example, pic001.jpg-pic100.jpg accorded.(2)We made a FlashPhotoPlayer by using Flash. This player can load images of specific paths by

using LoadMovie function, and the user can play or stop the player by clicking available button.(3)We stored FlashPhotoPlayer to each author' folder

Because pictures' name is same in each folder, we just make a Flash player to load pictures in different folders. Fig.03 shows

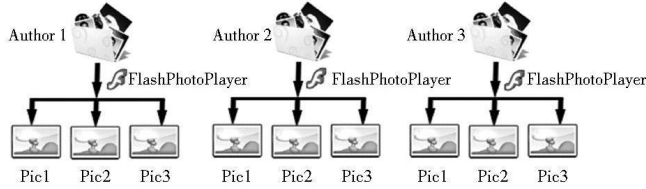


Figure 3. The location chart of FlashPhotoPlayer and image's storage location

In testing, we found in surprise that the button of saving picture didn't appear, and the file being downloaded by software such as Flashget was only FlashPhotoPlayer. The users can't download the picture. The effect of protection was very good. At the same time, the pictures could zoom in or zoom out and show their notes.

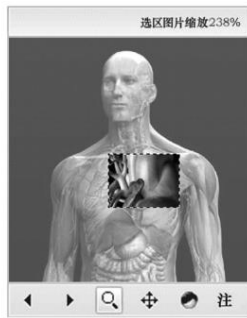


Figure 4. Using Flash to make images Player, the medical image can zoom in or zoom out

IV. HIDING CONFIDENTIAL PICTURES BY USING MEMBERSHIP SYSTEM

Website separates technically confidential pictures and ordinary users by using membership. This is an effective method.

Under the support of dynamic web pages and database technology, the system what we designed realizes the following function.

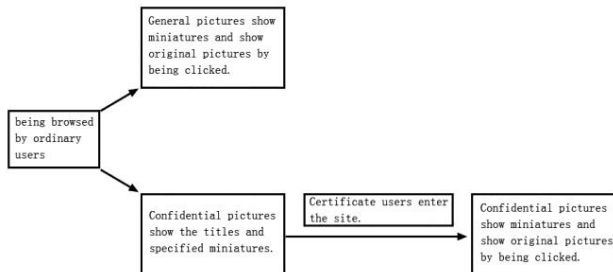


Figure 5. System design

In order to facilitate the management, the users are divided into five levels according to access and downloading right, and we set the right for each level (in Table 3).

TABLE 3. MEMBERSHIP'S LEVELS AND PERMISSIONS SETTINGS

Level	Right
administrator	All columns can be browsed and uploaded, downloaded.
officer	All columns can be browsed and downloaded.
Teachers user	Unconfidential pictures can be browsed and downloaded.
Students user	Unconfidential pictures can be only browsed.
General visitors	Ordinary pictures can be only browsed.

After the registration by user, we can give the user the corresponding right in the background according to the true name and place of work. After the user enter the site, the site will selectively display confidential images at the front desk according to the user's right. Thus the website images copyright can be well protected.

V. CONCLUSION

Today, it is a kind of trend that digital photography spreads in the network. The copyright protection of digital images has become an unavoidable issue under the network environment. If the copyright infringement problems can not be solved and resources lose the protection, so the enthusiasm of resources builders will be certainly weakened greatly. It is vital to right protection that the infringer and the social public improve the protection awareness in order to promote healthy and rapid development of creating work including science, culture and art.

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